TECHNICAL ANALYSIS REPORT

Why a CIR-Trained Model Cannot Work on RGB Images

Regardless of Model Architecture, Training Strategy, or Dataset Size

Application: Slum Detection and Segmentation from Satellite Imagery
Location: Mumbai, Maharashtra, India | Dataset: Extracting Slums from Satellite Imagery
Images Analysed: 000000026.tif, tile_5_15.tif, satellite11__1_.tif

Executive Summary

This report explains in full technical detail why a deep learning segmentation model trained on Colour Infrared (CIR) satellite imagery is fundamentally incompatible with standard Red-Green-Blue (RGB) images — regardless of the model architecture used, the size of the training dataset, or the training strategy applied. The incompatibility is not a software bug or a hyperparameter problem. It is a physical and spectral incompatibility rooted in what each image channel actually measures.

The core problem: Both CIR and RGB images have three channels and identical data shapes (H, W, 3). A model receiving either type of image sees the same array dimensions and the same value range (0–255). It has no way to detect which type it is receiving. However, the spectral content stored in Channel 0 of a CIR image (Near-Infrared reflectance) is physically and statistically opposite to the spectral content stored in Channel 0 of an RGB image (Red reflectance). The result is that the model applies learned CIR patterns to completely different data producing predictions that are not merely inaccurate but systematically inverted.

Critical Finding

A CIR-trained slum detection model applied to RGB images will confidently misidentify vegetation as slum and slum rooftops as non-slum. This is not a small accuracy drop it is a fundamental inversion of learned knowledge. No amount of fine-tuning or threshold adjustment can fix this without changing the input data type.

1. Understanding CIR and RGB Imagery

1.1 The Electromagnetic Spectrum and Imaging

A camera whether on a phone, a drone, or a satellite records light by measuring how much electromagnetic radiation bounces off a surface and reaches the sensor. The human eye is sensitive to wavelengths between approximately 400 nanometres (violet) and 700 nanometres (deep red). Camera sensors can be built to detect any portion of this spectrum, including wavelengths the eye cannot see.

Satellite missions designed for land monitoring deliberately capture wavelengths beyond the visible range because invisible wavelengths carry information about surface materials that visible light cannot reveal. Near-Infrared (NIR) radiation, at wavelengths of approximately 750 to 900 nanometres, is the most important such band for distinguishing vegetation, urban surfaces, and bare soil.

1.2 RGB — What a Standard Camera Captures

An RGB image contains three channels corresponding to the three colour receptors in the human eye. Every pixel stores three values representing how much Red (approximately 620–700 nm), Green (approximately 495–570 nm), and Blue (approximately 450–495 nm) light was reflected from that surface. This is what a standard camera, phone, drone RGB sensor, or web map tile service delivers.

Channel	Wavelength	What It Measures
0 — Red	620–700 nm	Red light reflectance. High in red/brown surfaces, rusty metals, exposed soil.
1 — Green	495–570 nm	Green light reflectance. Vegetation reflects some green (why leaves look green). Urban surfaces reflect moderate green.
2 — Blue	450–495 nm	Blue light reflectance. High in clear water, sky-lit surfaces. Low in most urban materials.

1.3 CIR — What a Multispectral Satellite Captures

A CIR (Colour Infrared) image also has three channels, but they contain entirely different wavelength information. The sensor replaces the Blue channel with Near-Infrared, shifting the entire spectral window toward longer wavelengths. This is deliberately done to maximise the information available for land cover classification.

Channel	Wavelength	What It Measures
0 — NIR	750–900 nm	Near-Infrared reflectance. Invisible to humans. Strongly reflected by healthy vegetation cell walls. Absorbed by water.
1 — Red	620–700 nm	Red light reflectance. Same as RGB Channel 0 — but now in position 1.
2 — Green	495–570 nm	Green light reflectance. Same as RGB Channel 1 — but now in position 2.

Critical Observation

Both image types have exactly three channels. Both store uint8 values in the range 0–255. They are indistinguishable as NumPy arrays or PyTorch tensors. The model cannot tell them apart it simply reads three numbers per pixel and applies learned filters. This is precisely why the incompatibility is so dangerous: it fails silently.

1.4 Why Your Image Looks Red

When you opened 000000026.tif in a standard image viewer, it appeared with large bright red blobs covering the left half. This is because the viewer rendered Channel 0 as Red, Channel 1 as Green, and Channel 2 as Blue the standard RGB rendering assumption. But Channel 0 contains NIR data, not Red data. Healthy vegetation reflects NIR very strongly (reflectance of 40–60%), producing high Channel 0 values of 180–240. The viewer colours these high values as bright red, making all vegetated areas appear vivid red.

This is not a file corruption or display error. It is correct CIR data being interpreted with an RGB colour mapping. The actual scene contains dense tree canopy (left half, appearing red) and a dense urban settlement with slum characteristics (right half, appearing in mixed neutral colours).

2. The Physics of Near-Infrared and Why It Cannot Be Replaced

2.1 How Vegetation Interacts With Light

A plant leaf has a layered internal structure. The outermost layer, the epidermis, is largely transparent to all wavelengths. Below it is the mesophyll, a spongy tissue filled with air pockets and cell walls. This structure creates a highly effective reflector for NIR radiation through multiple internal scattering events, while simultaneously using the Red wavelengths for photosynthesis (chlorophyll absorbs Red strongly) and reflecting some Green (which is why leaves appear green to human eyes).

Wavelength Band	Interaction with Leaf	Reflectance %	Pixel Value (0–255)
Blue (450–495 nm)	Absorbed by chlorophyll and carotenoids	5–10%	13–26
Red (620–700 nm)	Strongly absorbed for photosynthesis	5–10%	13–26
Green (495–570 nm)	Partially reflected (gaps in chlorophyll absorption)	10–20%	26–51
NIR (750–900 nm)	Strongly reflected by mesophyll cell wall structure	40–60%	102–153

This creates an enormous spectral contrast between vegetation and non-vegetation surfaces that simply does not exist in the visible spectrum. In a CIR image, a tree with Channel 0 (NIR) value of 200 and Channel 1 (Red) value of 30 produces an NDVI of (200-30)/(200+30) = +0.74. A slum rooftop with Channel 0 value of 90 and Channel 1 value of 80 produces NDVI of (90-80)/(90+80) = +0.06. These values are clearly separated. In RGB, the Red channel of a tree (low, ~30) and the Red channel of a corrugated iron roof (high, ~160) appear at opposite ends, but the separation is less consistent and highly dependent on lighting, season, and roof material colour.

2.2 Why Slum Rooftops Have a Specific NIR Signature

Informal settlements in cities like Mumbai are dominated by corrugated galvanised iron (CGI) sheets, thin asbestos panels, and recycled plastic sheeting. These materials have characteristic NIR reflectance properties that differ from both vegetation and formal construction materials such as reinforced concrete and clay tiles.

Surface	NIR Ch0	Red Ch1	NDVI	Identifiable from CIR?
Dense Vegetation	180–240	20–40	+0.64 to +0.84	YES very high NIR
Slum CGI Rooftop	80–120	70–110	+0.03 to +0.18	YES medium NIR, low NDVI
Formal Concrete Roof	100–140	90–130	+0.04 to +0.12	PARTIAL similar to slum
Bare Soil / Road	60–100	80–130	-0.05 to +0.08	YES low NIR relative to Red

Water Body	5–30	20–60	-0.4 to -0.1	YES NIR strongly absorbed

In the RGB domain, a corrugated iron roof (rusty reddish-brown) has a high Red channel value of 150–200. A dense tree canopy has a low Red channel value of 30–60. But so does a shaded road or a dark water tank. The RGB separability between slum rooftops and other urban materials is much weaker than the CIR separability. This is why CIR was chosen for this dataset in the first place.

3. The Channel Inversion Problem The Core Incompatibility

3.1 What the Model Learns From CIR Training Data

During training, the FPN-VGG16 or any model processes hundreds of CIR image tiles and their corresponding binary slum masks. The convolutional filters in the first VGG16 layer learn to respond to specific combinations of values across the three input channels. The most important patterns learned are those that distinguish slum pixels from non-slum pixels. Because Channel 0 contains NIR data throughout all training images, the model implicitly learns rules about NIR reflectance.

The key learned associations in the model's first layers are approximately:

- **HIGH Channel 0 (NIR ~180–240) + LOW Channel 1 (Red ~20–40):** This combination has very high NDVI. The model learns: this is vegetation. Label is 0 (not slum). Do NOT activate slum prediction.
- **MEDIUM Channel 0 (NIR ~80–120) + MEDIUM Channel 1 (Red ~70–110):** Moderate NIR, balanced with Red. The model learns: this is a dense rooftop cluster. Check spatial context. Likely slum.
- **LOW Channel 0 (NIR ~5–30) across all channels:** NIR strongly absorbed. The model learns: this is water or deep shadow. Label is 0 (not slum).
- **UNIFORM LOW-MEDIUM all channels:** Smooth reflectance. The model learns: paved road or open ground. Not slum.

3.2 What Happens When an RGB Image Is Fed to the Same Model

When an RGB image is passed through the model, Channel 0 now contains Red reflectance instead of NIR reflectance. The model applies the same filters it learned for NIR data, but the physical meaning of every value has changed completely.

Surface Type	Ch0 in CIR (NIR)	Ch0 in RGB (Red)	Model Decision on CIR	Model Decision on RGB
Dense Vegetation (Trees)	HIGH: 200	LOW: 35	Correct: NOT slum (NDVI=+0.74)	WRONG: model sees low Ch0 = could be slum
Slum CGI Rooftops	MEDIUM: 95	HIGH: 170 (rusty metal)	Correct: SLUM (low NDVI, dense pattern)	WRONG: model sees high Ch0 = vegetation = NOT slum
Bare Soil / Dirt Road	LOW-MED: 80	MED-HIGH: 130 (brown tones)	Correct: NOT slum (no rooftop texture)	UNRELIABLE: high Ch0 = vegetation signal
Water Body	VERY LOW: 15	LOW-MED: 50 (blue water)	Correct: NOT slum (NIR absorbed)	PARTIAL: low value still signals non-slum, but unreliable
Formal Concrete Roof	MED: 120	MED: 110 (grey concrete)	Partial: similar to slum, relies on spatial context	PARTIAL: grey values overlap — unreliable

3.3 Why This Cannot Be Fixed by Adjusting the Threshold

A common instinct when a model performs poorly is to lower or raise the prediction threshold (the probability cutoff above which a pixel is classified as slum). This is not a valid fix for the CIR-RGB incompatibility. The problem is not that the model is predicting probabilities of 0.48 when it should be 0.52 (a threshold problem). The problem is that the model is predicting high slum probability for pixels that are actually trees, and low slum probability for pixels that are actually slum rooftops. No threshold value can correct a systematic feature-level inversion.

3.4 Why This Cannot Be Fixed by Using a Better Architecture

The architecture of the model — whether FPN, U-Net, DeepLab, SegFormer, or any other does not affect this incompatibility. All deep learning segmentation architectures process input channels by learning filters that respond to specific value patterns per channel. If the physical meaning of Channel 0 changes between training and inference, all architectures are equally affected. A more powerful architecture will simply learn the wrong CIR-specific patterns more accurately and fail more confidently on RGB images.

3.5 Why This Cannot Be Fixed by Training on More CIR Data

Collecting more CIR training images and labels improves the model's performance on CIR inference images. It does not help with RGB inference at all. The fundamental spectral mismatch between CIR Channel 0 (NIR) and RGB Channel 0 (Red) is unchanged regardless of dataset size. A model trained on one million CIR images will fail just as completely on an RGB image as a model trained on one hundred CIR images because the channel content mismatch is not a generalisation problem, it is a domain definition problem.

4. Three Satellite Images — Full Analysis

Three TIFF files were analysed during this project. Their physical properties determine whether the trained model can use them reliably.

4.1 000000026.tif — CIR Training Type Image

Property	Value and Meaning
Array Shape	(400, 400, 3) 400 rows × 400 columns × 3 bands
Data Type	uint8 values from 0 to 255 per channel per pixel
Band Content	Channel 0 = NIR, Channel 1 = Red, Channel 2 = Green — CIR order
Channel 0 (NIR) Mean	79.6 moderate NIR indicating mixed urban/vegetation scene
Channel 1 (Red) Mean	73.2 consistent with urban surfaces
Channel 2 (Green) Mean	81.4 moderate green reflectance
Pixel Resolution	0.8 m per pixel a 3-metre slum dwelling occupies approximately 4 pixels
Coordinate System	WGS 1984 UTM Zone 43N (EPSG:32643) covers central India, equal-area projection
Geographic Location	Approximately 19.04°N, 72.85°E Mumbai area, Maharashtra, India
Compression	LZW lossless, no pixel value degradation from compression
Visual Appearance	Bright red blobs (vegetation) on left, mixed neutral tones (urban settlement) on right. Diagonal road/railway divides the scene.
Model Compatibility	FULLY COMPATIBLE this is the training distribution

4.2 tile_5_15.tif — Mosaic Tile (RGBA)

This image is tile row 5, column 15 from a larger orthophoto mosaic. The '4' in its 4-channel structure includes an Alpha transparency channel, which marks valid data areas versus edge gaps between tiles.

Property	Value and Meaning
Array Shape	(600, 600, 4) 4 channels including Alpha
Data Type	uint8 values 0 to 255
Band Content	Channel 0=Red, Channel 1=Green, Channel 2=Blue, Channel 3=Alpha (255 everywhere = fully opaque)
Channel 0 (Red) Mean	136.4 significantly higher than training Channel 0 (NIR mean 79.6)
Channel 1 (Green) Mean	135.9
Channel 2 (Blue) Mean	141.4 Blue channel not present in training data at all
Pixel Resolution	0.565 m per pixel higher resolution than training, so features appear proportionally larger

Coordinate System	WGS 84 UTM Zone 43N same projection as training data
What Model Receives	After src.read()[:3]: Channel 0=Red (not NIR), Channel 1=Green, Channel 2=Blue. Alpha is silently dropped.
Model Compatibility	NOT COMPATIBLE — RGB not CIR. Channel 0 meaning is inverted. Spectral signatures are wrong.

4.3 satellite11__1_.tif — Web Map Tile (RGB)(samGEO)

This image was downloaded from a web mapping service (Google Maps, Bing Maps, or similar). Web map tiles are always RGB, always compressed with JPEG at some point in their pipeline, always use Web Mercator projection, and never contain NIR data.

Property	Value and Meaning
Array Shape	(769, 1118, 3) wider area, non-square tile
Data Type	uint8 values 0 to 255
Band Content	Channel 0=Red, Channel 1=Green, Channel 2=Blue standard RGB
Channel 0 (Red) Mean	97.3 similar range to training Channel 0 (NIR mean 79.6) by coincidence, but physically different
NIR Band	NOT AVAILABLE and NEVER AVAILABLE — web map services do not include NIR
Pixel Resolution	1.19 m per pixel coarser than training data, features appear smaller after resize
Coordinate System	WGS 84 Pseudo-Mercator (EPSG:3857) Web Mercator projection, different from training UTM
Compression Artefacts	JPEG compression applied at tile serving. Introduces block artefacts that alter pixel values by ±3 to ±15 units.
Model Compatibility	NOT COMPATIBLE RGB, no NIR, wrong projection, coarser resolution, JPEG artefacts

5. Additional Incompatibilities Beyond Band Content

Even if the band content problem were somehow resolved, three further incompatibilities exist between the CIR training data and the RGB test images.

5.1 Spatial Resolution Mismatch

The model was trained on images where each pixel represents 0.8 metres of ground. At this scale, the characteristic texture of a slum densely packed small dwellings with irregular spacing, narrow lanes between structures produces a specific spatial frequency pattern in the image. The model's convolutional filters learned to respond to this pattern at the scale of approximately 4–10 pixels per structure.

Image	Resolution	3m Dwelling (pixels)	Effect
Training data (000000026)	0.80 m/px	~4 px	Reference scale
tile_5_15	0.565 m/px	~5–6 px	Features 40% larger
satellite11	1.19 m/px	~2–3 px	Features 50% smaller

After resizing all images to 256×256 pixels for model input, a slum dwelling that occupied 4 pixels at training resolution now occupies proportionally more or fewer pixels in inference images. The spatial frequency of the characteristic slum texture changes, reducing the model's ability to recognise the pattern it learned.

5.2 Coordinate System (Projection) Mismatch

The training images use UTM Zone 43N projection (EPSG:32643), which is an equal-area projection that accurately preserves distances and shapes across the image. satellite11 uses Web Mercator (EPSG:3857), used by Google Maps and OpenStreetMap. At Mumbai's latitude of approximately 19 degrees North, Web Mercator introduces a scale factor of approximately 1.006, meaning shapes are stretched by about 0.6% in both dimensions compared to reality. Horizontal features (roads, building footprints) appear very slightly wider than in UTM imagery. While this is a minor effect at low latitudes, it is additive to the other incompatibilities.

5.3 Image Compression Artefacts

The training CIR images are stored with LZW compression, which is mathematically lossless every pixel value is preserved exactly as recorded by the sensor. The satellite11 web map (SAMGEO)tile was served through a JPEG compression pipeline before being saved as a TIFF. JPEG compression divides the image into 8×8 pixel blocks and approximates each block's frequency content, introducing quantization errors of typically ±3 to ±15 pixel value units. These artefacts create artificial block-shaped patterns at 8-pixel intervals throughout the image, which the model's convolutional filters will respond to as if they were real surface features.

5.4 Statistical Distribution Shift

Even if two images contained the same physical wavelengths, atmospheric conditions, sensor calibration, and image processing pipelines differ between image sources. The training data was captured by a specific aerial sensor calibrated for a specific atmospheric correction model. Web map tiles aggregate images from multiple sources and apply proprietary colour balancing and histogram normalisation. The resulting statistical distribution of pixel values across the three channels is fundamentally different from the training distribution. Models trained to minimise loss on one distribution will generally produce poor calibrated probabilities when applied to a different distribution.

6. Consequences of Applying a CIR Model to RGB Images

6.1 Prediction Quality

Metric	CIR model on CIR data	CIR model on RGB data	Change
F1 Score	~0.75–0.82	~0.20–0.40	Loss of 0.40+
IoU	~0.62–0.70	~0.12–0.25	Severe drop
Precision	~0.78–0.85	~0.15–0.35	Most predictions false
Recall	~0.72–0.82	~0.25–0.50	Misses true slum
False Positive Pattern	Rare, near slum boundaries	All vegetated areas predicted as slum	Systematic error
False Negative Pattern	Some boundary pixels missed	Large portions of actual slum missed	Systematic error

6.2 What You Would See in Practice

If you ran the trained CIR model on tile_5_15 (RGB) and visualised the prediction:

- The park or garden area in the tile would be predicted as slum — trees have high Red channel values in RGB, which the model interprets as the medium NIR signature of slum rooftops.
- The actual dense settlement area might be predicted as non-slum — corrugated iron rooftops have high Red values in RGB, which the model interprets as high NIR (vegetation signature = not slum).
- The prediction mask would look credible at first glance — it would contain coherent spatial regions, not random noise — making the error harder to detect without ground truth comparison.
- Adjusting the probability threshold from 0.5 to any other value would shift which surface types are included but would not correct the underlying inversion.

Dangerous Failure Mode

The model will not crash or produce obviously random outputs. It will produce smooth, spatially coherent predictions that look plausible. This is a silent failure — the predictions appear reasonable but are systematically wrong. This type of failure is particularly dangerous in applied settings where visual plausibility is mistaken for accuracy.

7. Solutions — What Can Be Done

There are three viable approaches. They differ in the trade-off between implementation complexity and prediction quality on RGB imagery.

Option A — Keep the CIR Model, Only Use It on CIR Images

The simplest approach. Do not attempt to use the trained model on RGB images. Only run inference on images from the same sensor and band configuration as the training data. Use tile_5_15 and satellite11 only for visual reference, not for model inference.

> **When to Choose This Option**
> Choose Option A if all your deployment images will come from the same aerial survey dataset as your training data. This is the most common scenario for research or municipal planning projects using a fixed data provider.

Option B — Convert CIR to Synthetic RGB, Retrain, Deploy on Real RGB

Generate approximate RGB images from your CIR data by extracting the Red and Green bands (which are present in CIR) and synthesising a Blue channel from the available data. Train a new model on these synthetic RGB images with the same binary slum labels. At inference, feed real RGB images — the spectral gap between synthetic Blue and real Blue is small in urban areas.

The synthesis formula used in this project:

```
Blue_synthetic = (Green × 0.70 + NIR × 0.10).clip(0, 255)
RGB_synthetic = stack([Red, Green, Blue_synthetic])
```

- **Advantage:** No new labelled data required. Same labels, same architecture. Change is only in the Dataset preprocessing step.
- **Advantage:** Retains ImageNet compatibility — VGG16 was pretrained on RGB images. Synthetic RGB is a better match to ImageNet statistics than CIR.
- **Limitation:** NIR information is discarded during training. The model can no longer use NIR-based vegetation separation. This may reduce F1 by approximately 0.05 to 0.12 points on CIR images.
- **Limitation:** Synthetic Blue is an approximation. In areas where real Blue and Green channels differ significantly (clear water, blue-painted roofs), the approximation degrades.

> **Expected Performance**
> A model trained on synthetic RGB from CIR labels and deployed on real RGB inference images can be expected to achieve approximately F1 0.65–0.72, compared to F1 0.75–0.82 for the CIR model on CIR images. This is a meaningful trade-off for substantially broader deployment.

Option C — Four-Channel Model Preserving NIR

Retain all four spectral values (Red, Green, synthetic Blue, NIR) and modify the VGG16 encoder to accept four input channels. This preserves the NIR signal for training while also being compatible with synthetic RGB inputs. The encoder weights are initialised by transplanting ImageNet pretrained weights for the first three channels and copying one existing channel for the fourth.

- **Advantage:** Highest accuracy on CIR data — NIR is preserved as an explicit channel. Expected F1 on CIR images: 0.78–0.85.

- **Advantage:** Slightly better than Option B on images where NIR is available, even if acquired from a different source.
- **Limitation:** Requires four channels at inference. Not compatible with standard RGB web tiles or phone camera images without acquiring NIR separately.
- **Limitation:** Loses ImageNet pretraining benefit for the fourth channel — it is initialised from a copy of an existing channel, not pretrained.

Summary Comparison of All Options

	Option A (CIR only)	Option B (Synth RGB)	Option C (4-channel)	
Works on CIR training data	YES	REDUCED	YES	Accuracy on same-source data
Works on tile_5_15 (RGBA)	NO	YES	NO (needs NIR)	
Works on satellite11 (RGB web tile)	NO	YES	NO (needs NIR)	
Works on phone/drone RGB	NO	PARTIAL	NO (needs NIR)	
New labelled data needed?	NO	NO	NO	
Implementation complexity	None	Low	Medium	
Expected F1 on CIR data	0.75–0.82	0.68–0.76	0.78–0.85	
Expected F1 on RGB data	0.20–0.40	0.65–0.72	0.20–0.40	

8. Final Summary

Root Cause

Channel 0 of a CIR image contains Near-Infrared reflectance. Channel 0 of an RGB image contains Red reflectance. These two measurements are physically opposite in their response to the most important surface types for slum detection. A model trained on CIR learns that Channel 0 HIGH means vegetation (not slum) and Channel 0 MEDIUM means rooftops (potential slum). In RGB, Channel 0 HIGH means red metal rooftops (slum) and Channel 0 LOW means vegetation (not slum). The model will systematically predict slum where there are trees and not-slum where there are corrugated iron rooftops.

Why Architecture Does Not Matter

This incompatibility is independent of model architecture. U-Net, FPN, DeepLab, SegFormer, and any other segmentation model all read the three input channels and apply learned filters. If the channel content changes between training and inference, all models fail equally. A more powerful model learns the wrong patterns more accurately and fails more confidently.

Why More CIR Training Data Does Not Help

More CIR training images improve CIR inference performance only. They do not change the spectral content of RGB images or bring NIR data into the RGB domain. The problem is not generalisation capacity it is that the model and the inference images are in fundamentally different spectral spaces.

Recommended Path Forward

If you need the model to work on standard RGB satellite imagery and web map tiles, implement Option B: extract Red and Green from your CIR training images, synthesise Blue from Green and NIR, and retrain the same FPN-VGG16 architecture on synthetic RGB images with the same labels. This requires no new labelled data and only a small change to the Dataset preprocessing code. Expected F1 on RGB inference images: 0.65–0.72.